

Name: _____

Exam Style Questions

Sine Rule

Cosine Rule

Area of a Triangle (Sine)



Corbettmaths

Equipment needed: Ruler, Pencil, Calculator and Pen

Guidance

1. Read each question carefully before you begin answering it.
2. Check your answers seem right.
3. Always show your workings

Video Tutorial

www.corbettmaths.com/contents

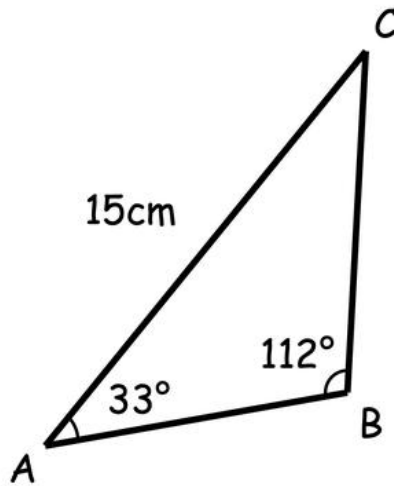
Videos 333 to 337



Answers and Video Solutions



1.

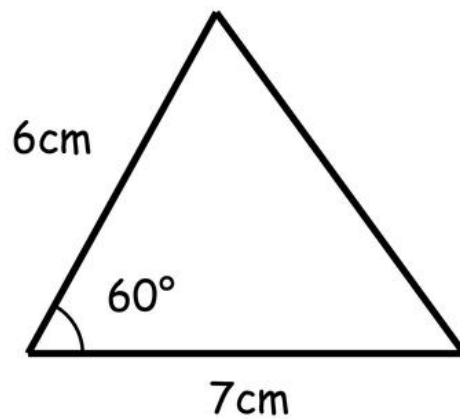


In triangle ABC the length of AC is 15cm
Angle ABC = 112°
Angle BAC = 33°

Work out the length of BC.

.....cm
(3)

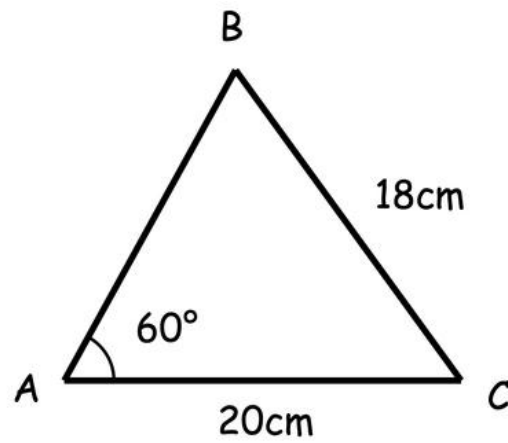
2.



Calculate the area of the triangle.

.....cm²
(2)

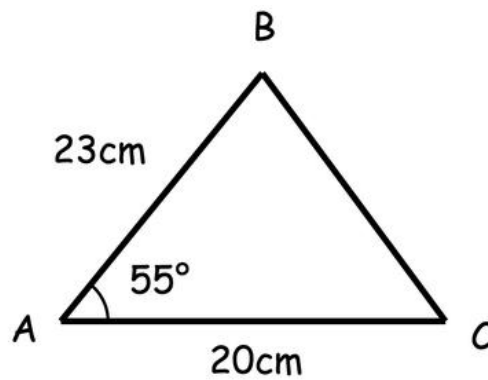
3.



Calculate the size of angle ABC.

.....°
(3)

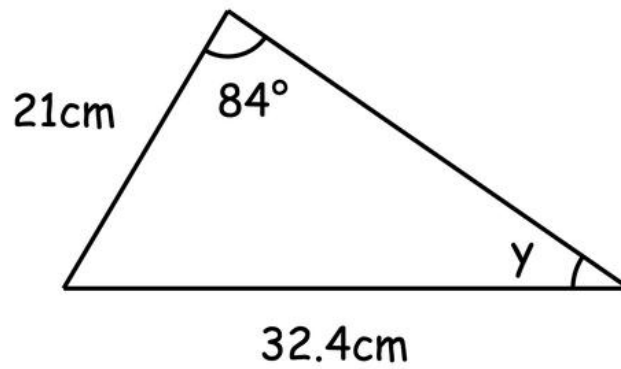
4.



Calculate the length of BC.

.....cm
(3)

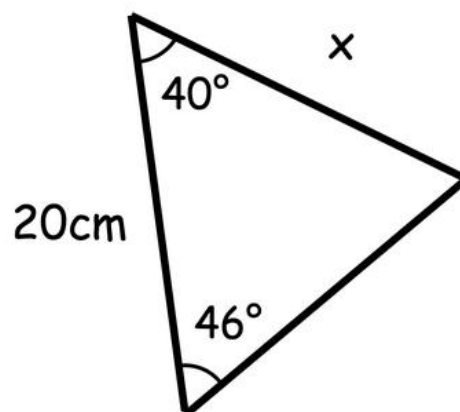
5.



Calculate the size of the angle labelled y .

.....^o
(3)

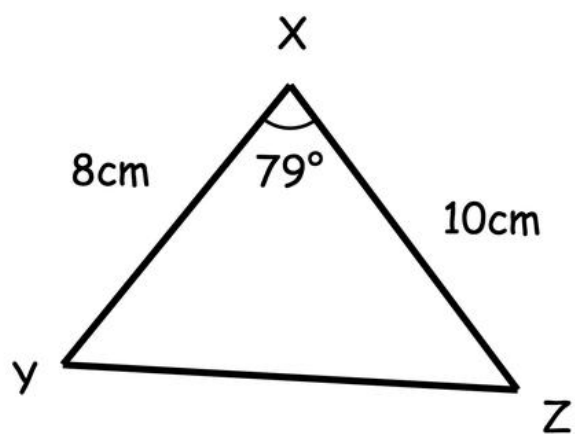
6.



Find the size of x .

.....cm
(3)

7.



$$XY = 8\text{cm}$$

$$XZ = 10\text{cm}$$

$$\text{angle } YXZ = 79^\circ$$

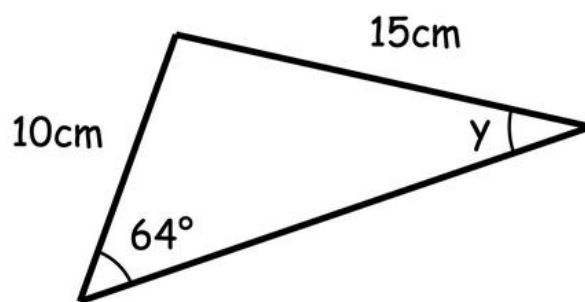
(a) Calculate the area of the triangle XYZ.

.....cm²
(3)

(b) Calculate the length of YZ.

.....cm
(3)

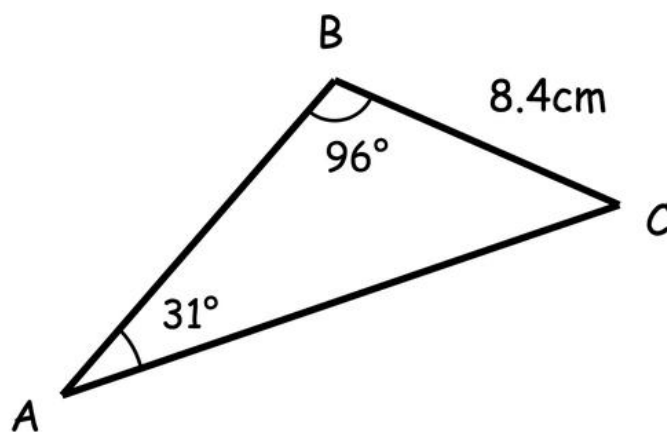
8.



Find the size of y .

.....^o
(3)

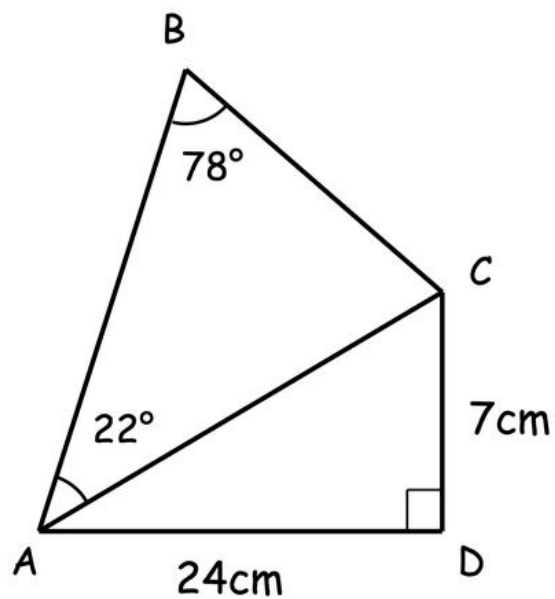
9.



Calculate the length of AB.

.....cm
(3)

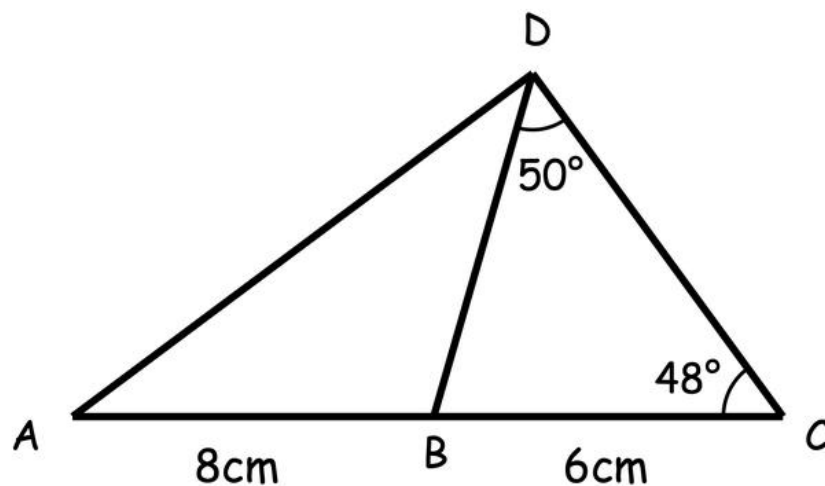
10. ABC and ACD are triangles



Calculate the length of BC.

.....cm
(4)

11.



ACD is a triangle and B is a point on AC.

AB = 8cm and BC is 6cm.

Angle BCD = 48° and angle BDC = 50° .

(a) Find the length of BD.

.....cm
(3)

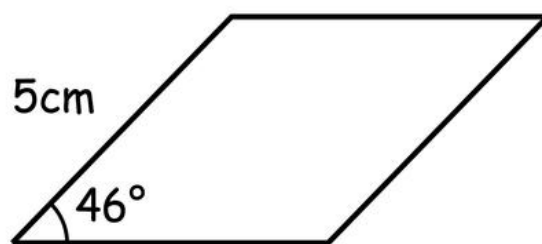
(b) Find the length of AD.

.....cm
(3)

(c) Find the area of triangle ABD.

.....cm²
(3)

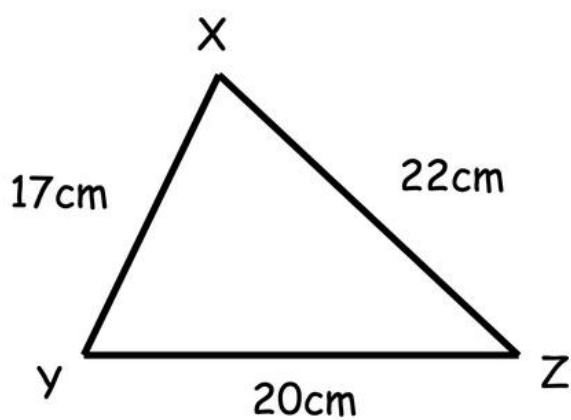
12. Shown below is a rhombus.



Find the area of the rhombus.

..... cm^2
(3)

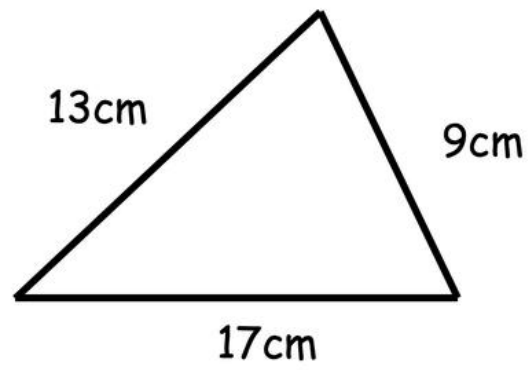
- 13.



Find the size of angle XZY.

..... $^\circ$
(3)

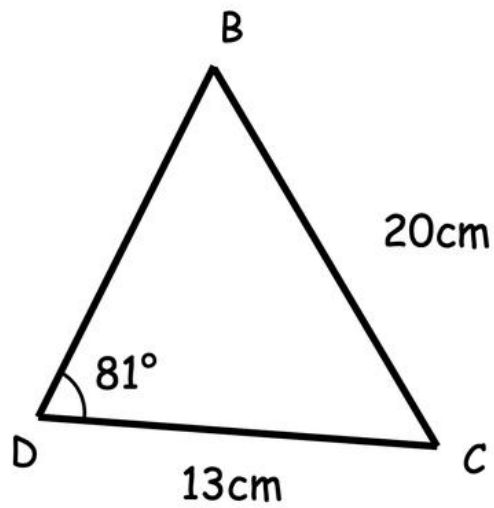
14.



Calculate the size of the smallest angle in the triangle.

.....°
(4)

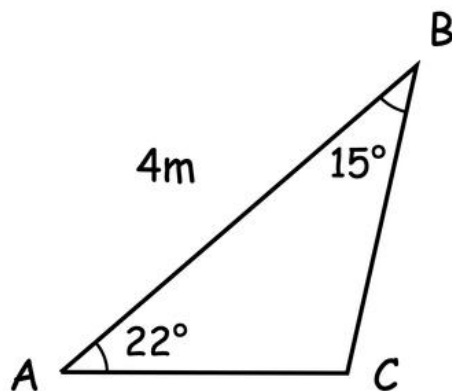
15. Shown below is triangle BCD.



Calculate the size of angle BCD.
Give your answer correct to 2 decimal places.

.....°
(4)

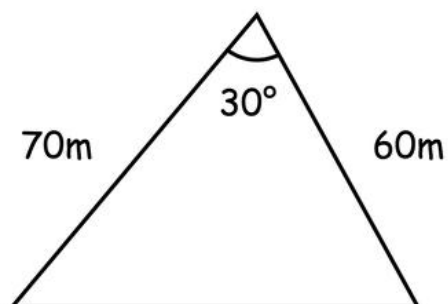
16. Shown below is triangle ABC.



Calculate the length of AC.

.....m
(3)

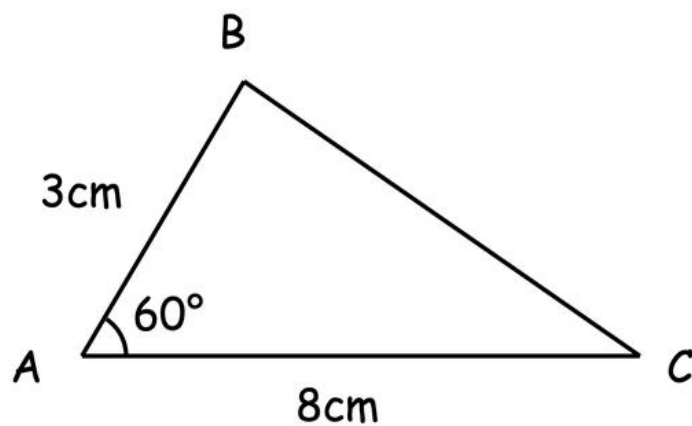
17. Shown below is a sketch of a field.



Work out the area of the field.

.....m²
(2)

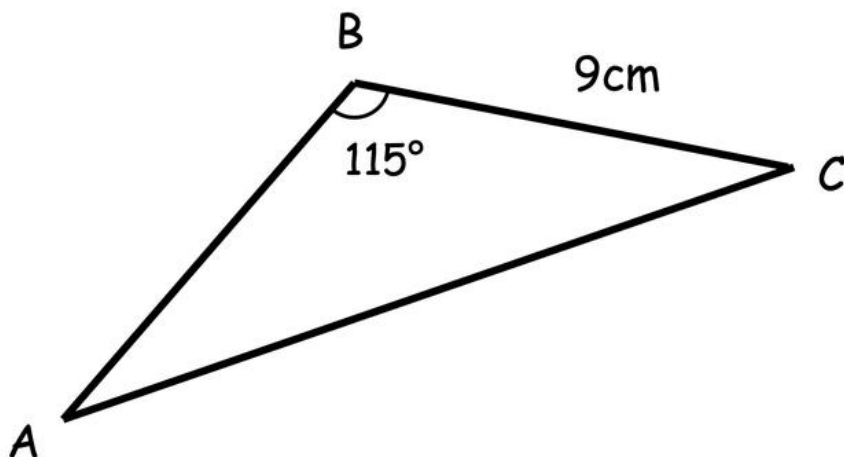
18. Shown below is triangle ABC.



Work out the length of BC.

.....cm
(3)

19. Shown below is triangle ABC.



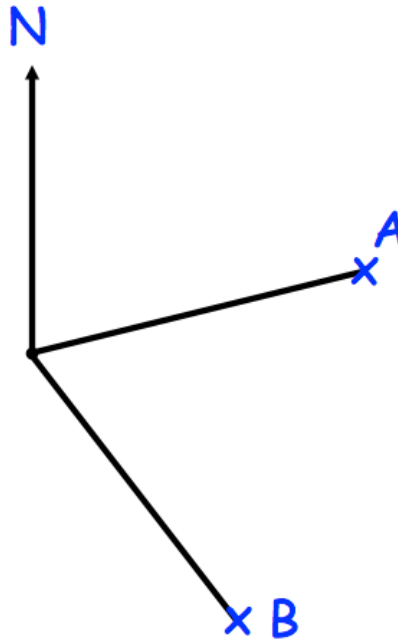
The area of triangle ABC is 72cm^2

Calculate the length of AB.

Give your answer to 2 decimal places.

.....cm
(3)

20.



Two ships, A and B, leave a port at 11:00

A travels on a bearing of 080° at a speed of 25km/h

B travels on a bearing of 152° at a speed of 20km/h

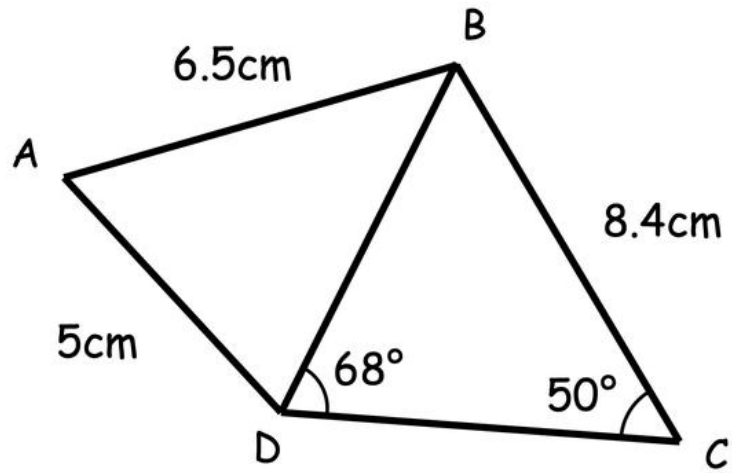
(a) Work out the distance between A and B at 14:00

.....km
(3)

(b) Work out the bearing of B from A at 14:00

..... $^\circ$
(3)

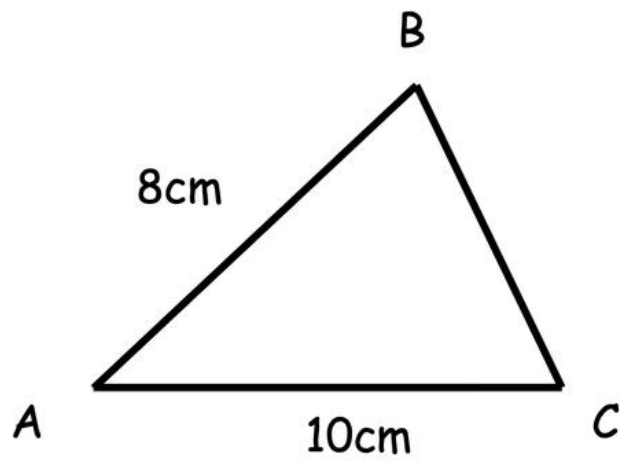
21.



Calculate the size of angle ABD.

.....°
(4)

22.



$$AB = 8\text{cm}$$

$$AC = 10\text{cm}$$

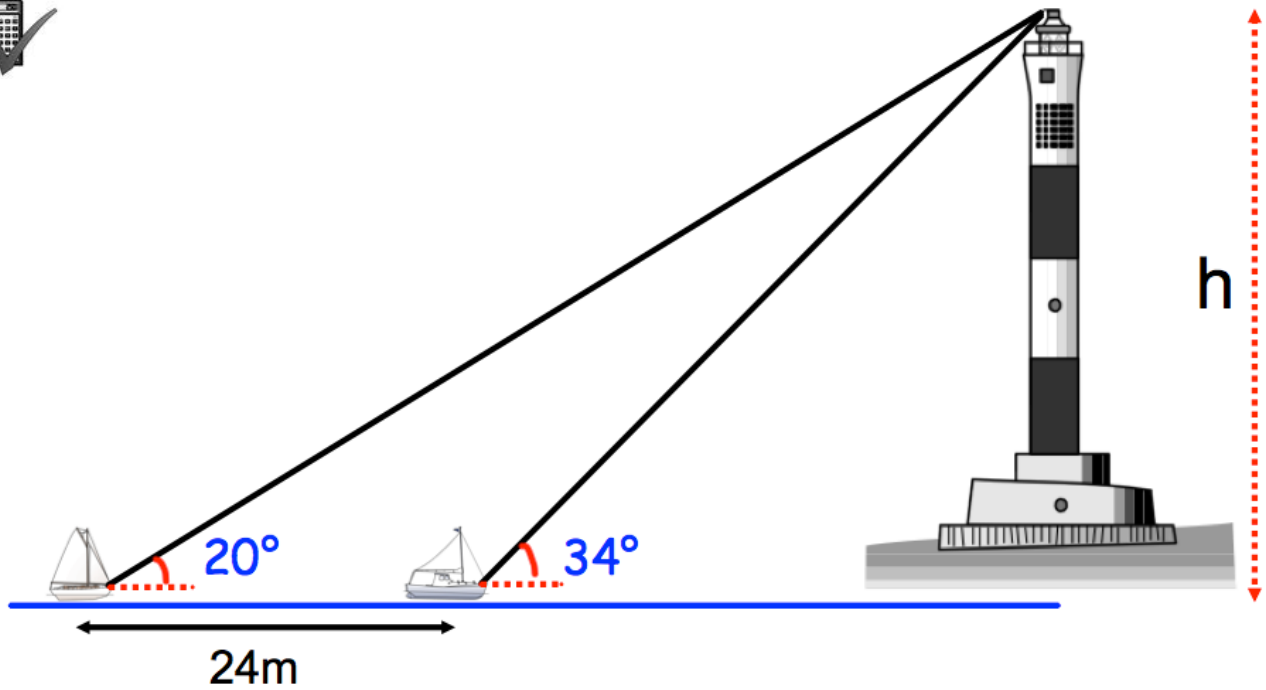
Angle BAC is acute.

The area of the triangle shown is 25cm^2

Calculate the perimeter of the triangle

.....cm
(4)

23.



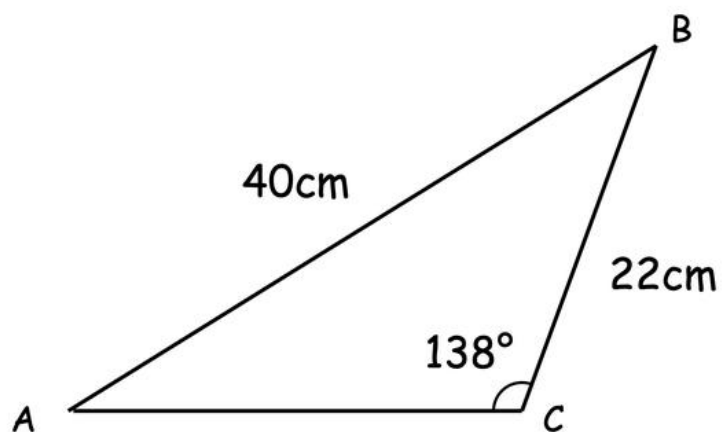
Two small boats are 24m apart.

The angle of elevation of the boats to the top of a lighthouse are 20° and 34° respectively.

Calculate the height of the lighthouse.

.....m
(6)

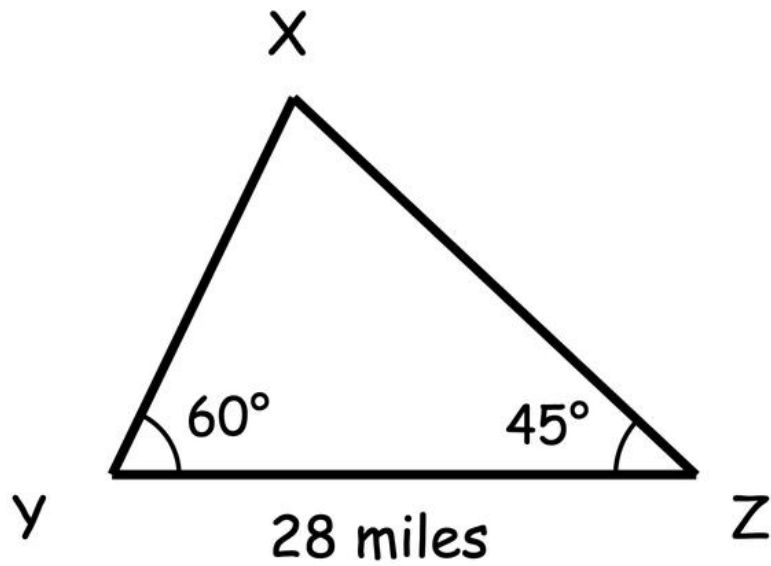
24. Shown below is triangle ABC



Calculate the length of AC.
Give your answer to 2 decimal places.

.....cm
(5)

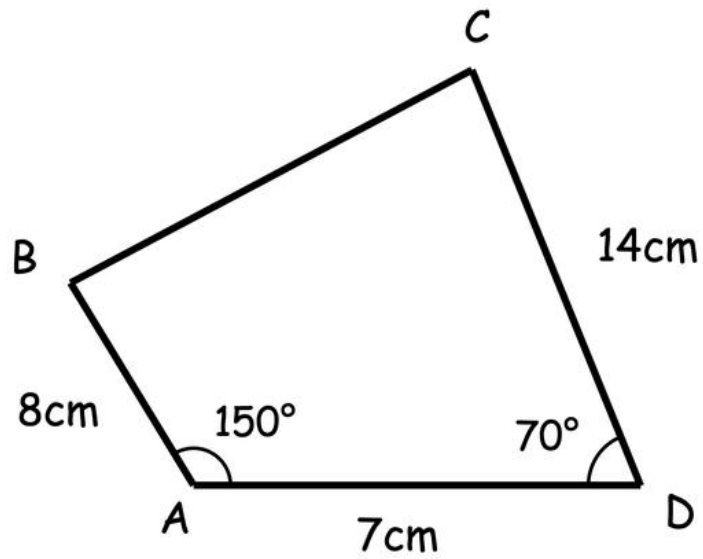
25. A boat, located at position X is running out of fuel.
There are two ports located at Y and Z.
The boat must refuel as soon as possible.



How much closer is the boat to the port at Y than the port at Z?

.....miles
(4)

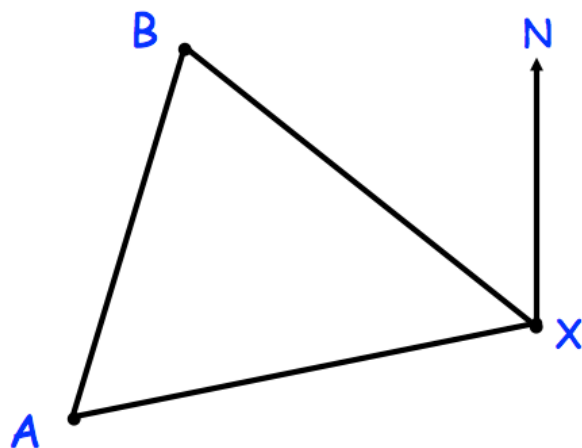
26. In a quadrilateral ABCD, $AD = 7\text{cm}$, $AB = 8\text{cm}$ and $CD = 14\text{cm}$.
Angle $BAD = 150^\circ$ and Angle $ADC = 70^\circ$



Calculate the length BC.

.....cm
(6)

27.



Ship A is 50km from X on a bearing of 258°
 Ship B is 44km from X on a bearing of 312°

(a) Calculate the distance between A and B.

.....km
(3)

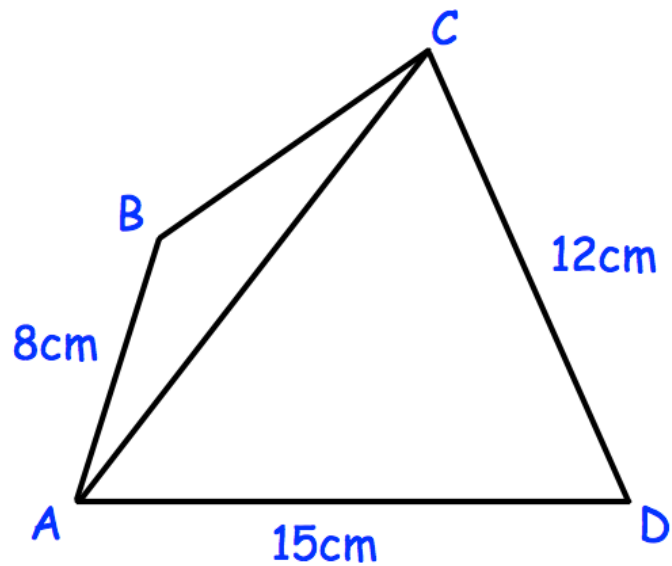
(b) Calculate the bearing of A from B.

..... $^\circ$
(3)

28. ABCD is a quadrilateral.



AB = 8cm, AD = 15cm and CD = 12cm.
Angle ADC = 78° and angle BAC = 20°



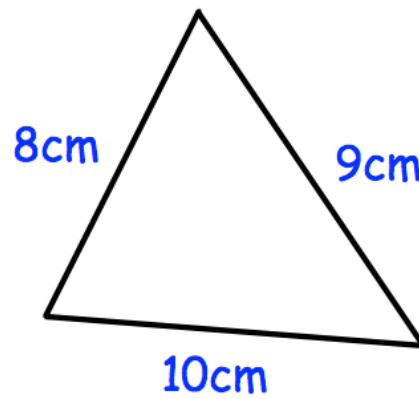
(a) Calculate the length of AC.

.....cm
(3)

(b) Calculate the area of triangle ABC.

.....cm²
(2)

29.



Find the area of the triangle.

.....cm²
(5)

30. A clock has two hands.

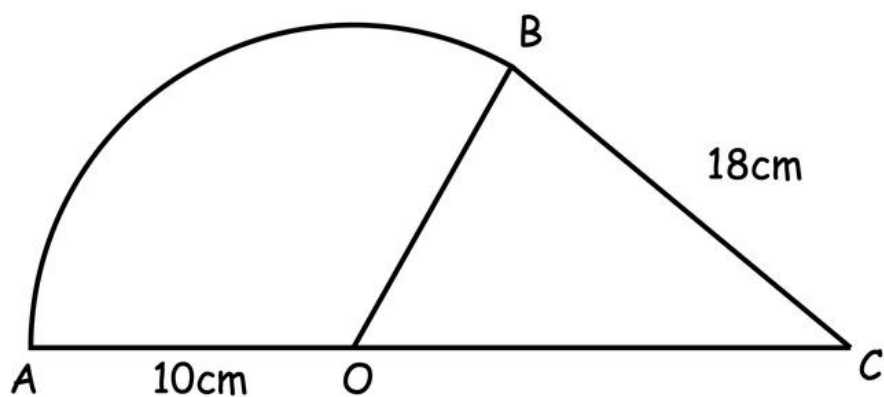


A minute hand that is 7cm long and an hour hand that is 5cm long.

Find the distance between the tips of the two hands at 13:40

.....cm
(4)

31. OAB is a sector of a circle, centre O.
AOC is a straight line.



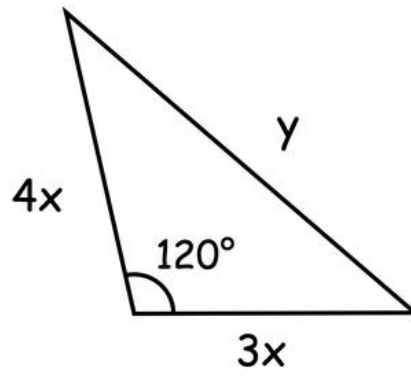
$$AO = 10\text{cm} \quad BC = 18\text{cm}$$

Arc AB has a length of 19.5cm

Calculate the size of angle OCB.

.....°
(6)

32.



Work out the ratio $y : x$

.....
(4)